Computer Networks I
Reti di Calcolatori I

Introduction to Computer Networks

Lecture 1

Università degli Studi di Napoli Federico II

Overview
Syllabus

- Introduction to computer networks: goals and applications, main notions, network topologies and types, services (ISPs, grid, cloud, IoT), standards (layered models, IOS/OSI and reference models, protocols).
- Application layer: client-server ad peer-to-peer architectures, FTP and SSH protocols (basis), structure of URL, HTTP protocol, mailing protocols (STMP, POP3, IMAP), DNS, BitTorrent protocol.
- Transport layer: fundamentals, UDP protocol, reliable data transfer, TCP protocol.
- Network layer: fundamentals, routers, IPv4 protocol, DHCP, NAT, IPv6, families of routing algorithms.
- Access layer (data link + physical): network interfaces, parity check and CRC, collisions, ARP protocol, Ethernet protocol (basis).
- Introduction to network security: attacks on networks, introduction to cryptography, message integrity, AP and TLS/SSL, operational security (firewall, IDS).
- Network programming: UDP and TCP sockets, REST applications, data exchange formats.

Uses of computer networks

Computer network

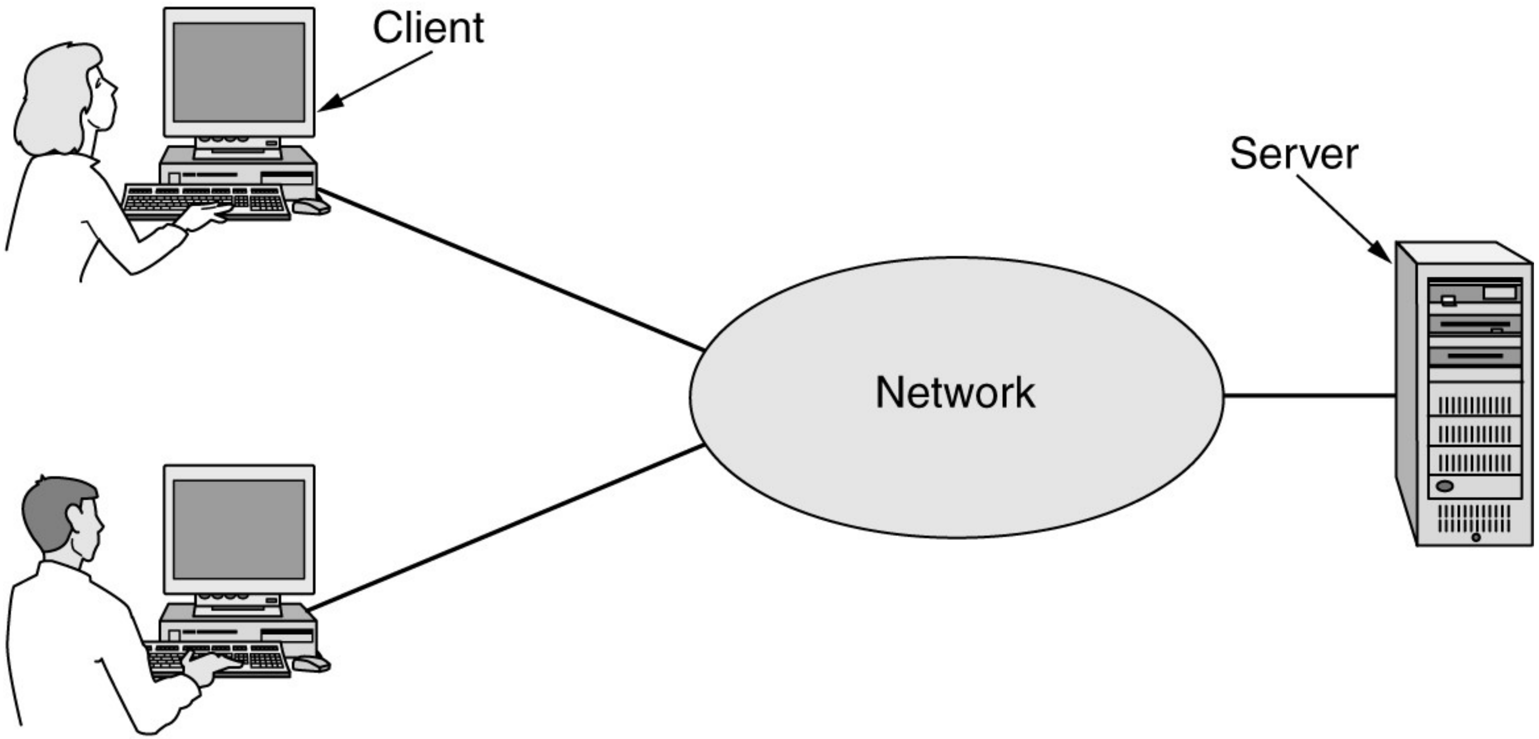
- Large number of separate but interconnected computers do a job
- Collection of interconnected, autonomous computing devices
- Interconnected computers can exchange information

Example: the Internet

Network uses

- Access to information
- Person-to-person communication
- Electronic commerce
- Entertainment
- The Internet of Things

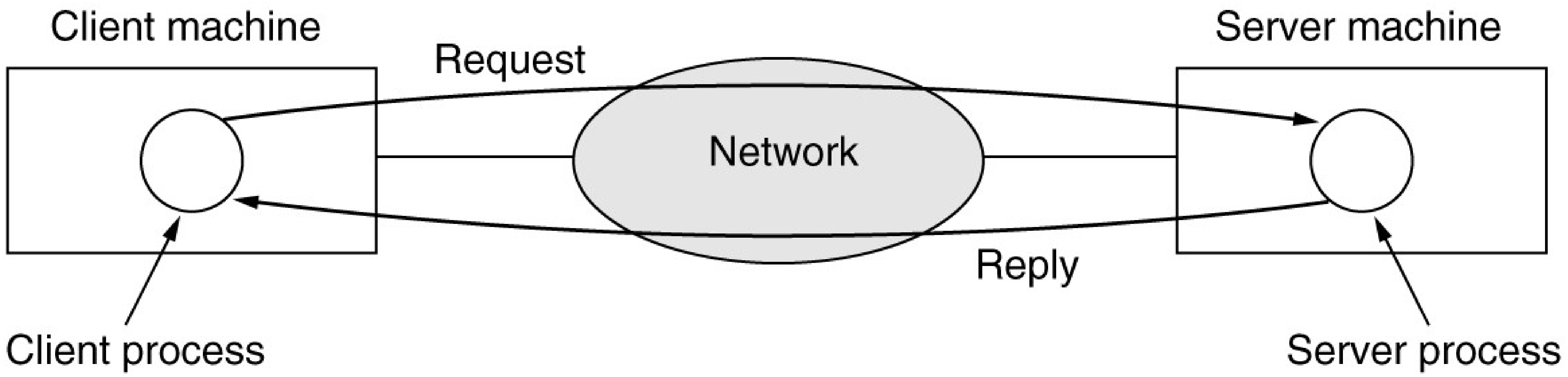
Access to Information



In the client-server model, a client explicitly requests information from a server that hosts that information.

Access to information

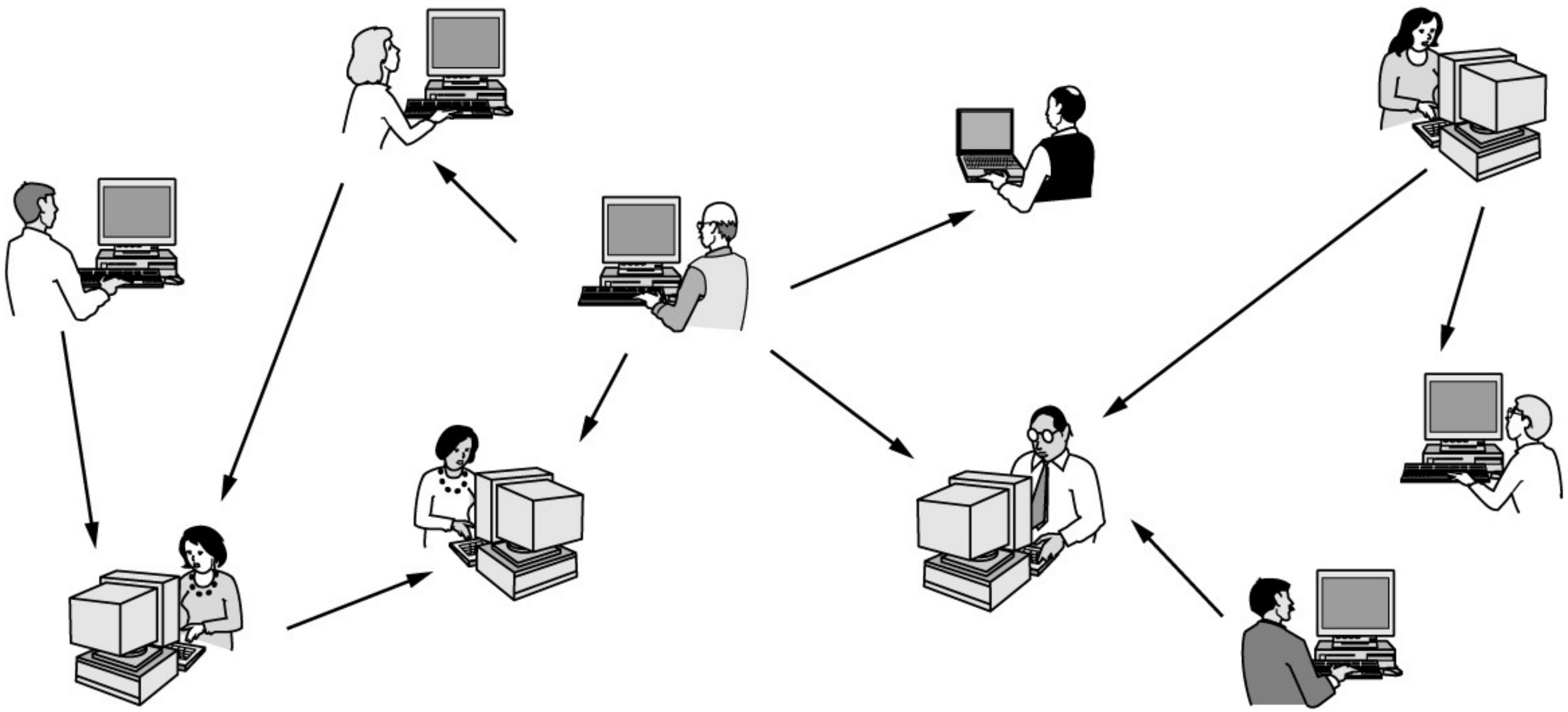
Client-server



Communication takes the form of the client process sending a message over the network to the server process. The client process then waits for a reply message.

Access to information

Peer-to-peer



In a peer-to-peer system, there are no fixed clients and servers.

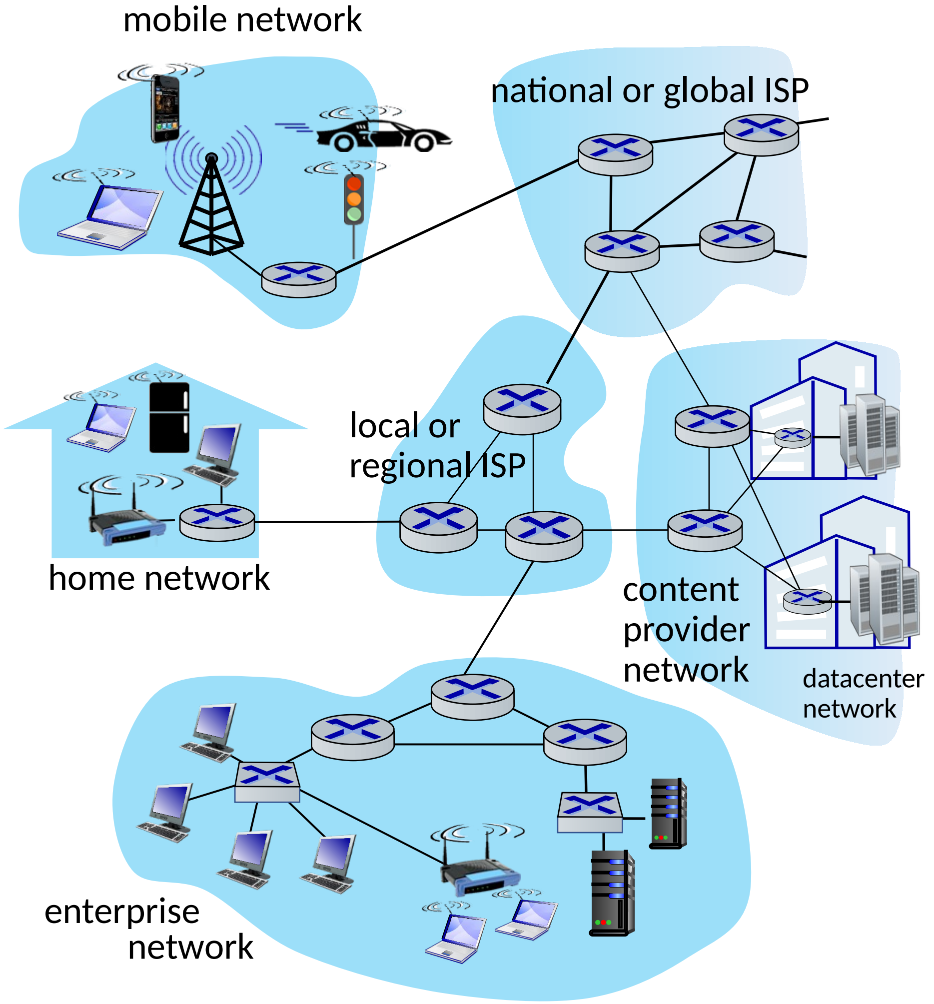
Introduction

Computer Networks and Internet

Computer networking: the process of connecting computer together so that they can share information.

[Cambridge Dictionary]

- **Internet** (the most important computer network in the world)
  - The Internet includes hundreds of millions of network **devices**, **links**, **computers**, etc. offering hundreds of **services** for the users.
- There are also **smaller networks** (local or detached from internet).



Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

Internet is a network of networks. It is technically a huge *Wide Area Network* (WAN) that allows sub-networks and devices to be connected over different nations and continents.

The idea of such "universal" big network was initially theorized by researchers (most notably J.C.R. Licklider) around 1950 and started to become reality almost 10 years later.

Introduction
History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).

Initial ARPANET having 3+1 nodes: SRI, UCLA, UCSB, and UTAH (1969)

The initial ARPANET was mainly a closed network designed to include universities and research centers. Similar networks were independently created all around US and Europe.

Introduction

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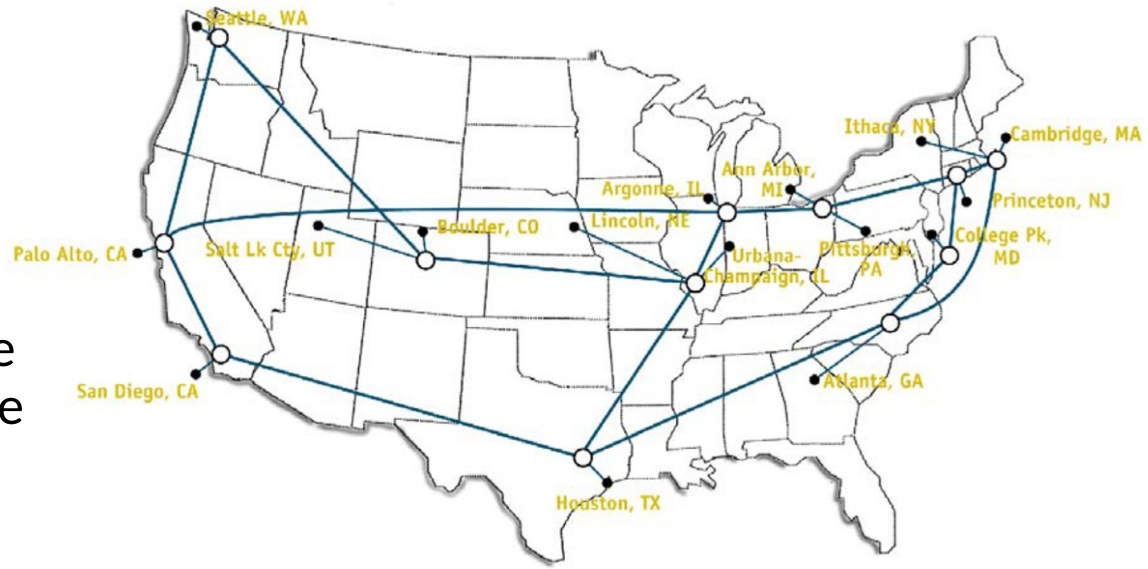
  - R. Kahn (ARPA) and V. Cerf (Stanford) designed the Transmission Control Protocol (TCP) and Internet Protocol (IP), two protocols of the **Internet protocol suite** (1974).
  - The National Science Foundation (NSF) awarded contracts for the **NSFNET** project, a TCP-IP based network (1986).



NSFNET in 1992 (Wikipedia).

The NSFNET and the TCP/IP protocol went toward the idea of a *network of networks*, but both ARPANET and NSFNET carried **commercial restrictions**.

Introduction

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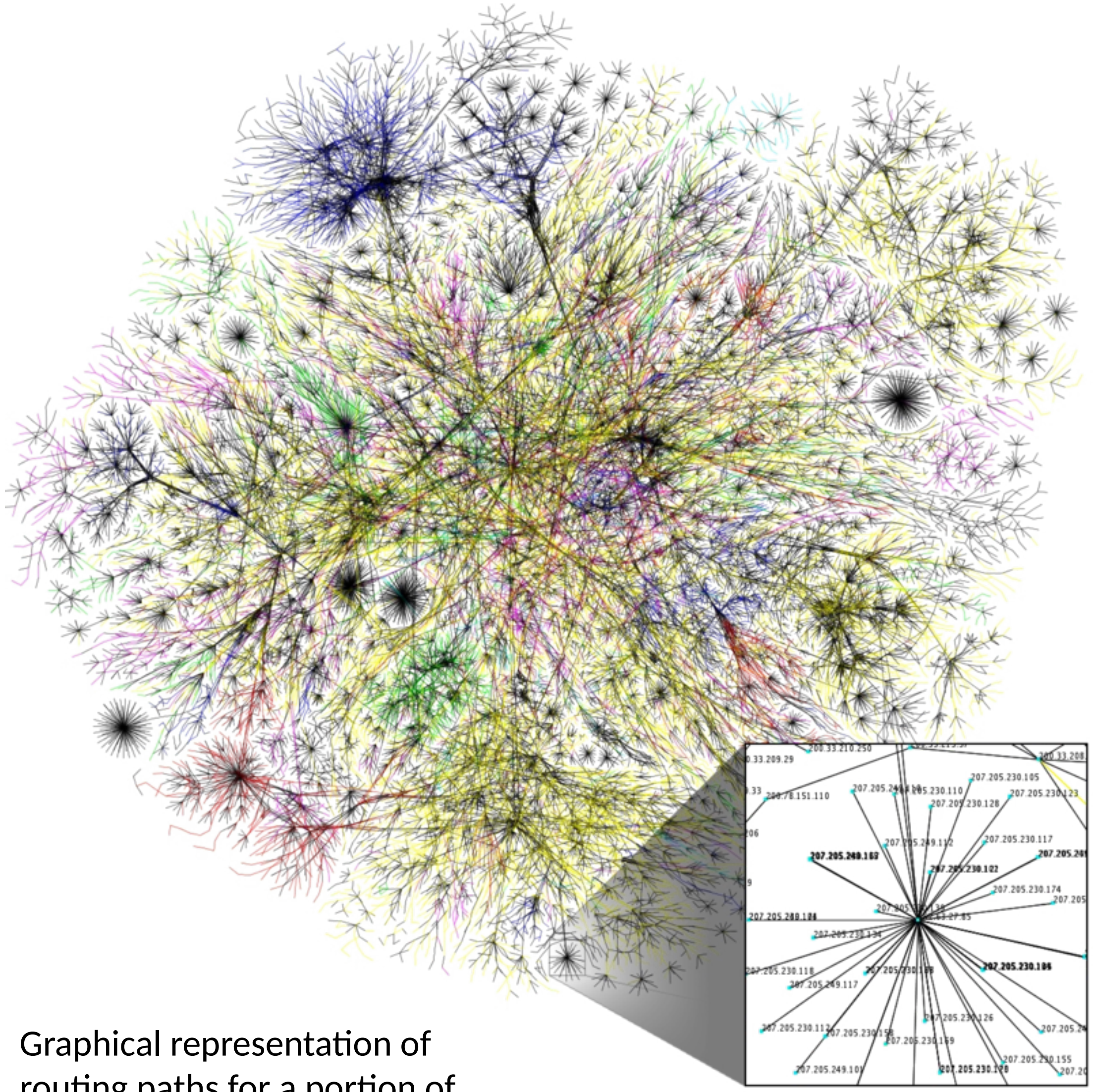
  - The **ARPANET decommissioned** (1990).
  - The **NSFNET decommissioned**, removing the last restrictions on the use of the Internet for commercial traffic (1995).

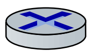


Graphical representation of routing paths for a portion of Internet (Wikipedia).

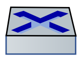
Introduction

Networks Components: Devices, Links, Hosts

- Network **devices** and **links** are the **infrastructure** that allow hosts to connect:


Router

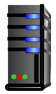

Access Point


Switch


Hub

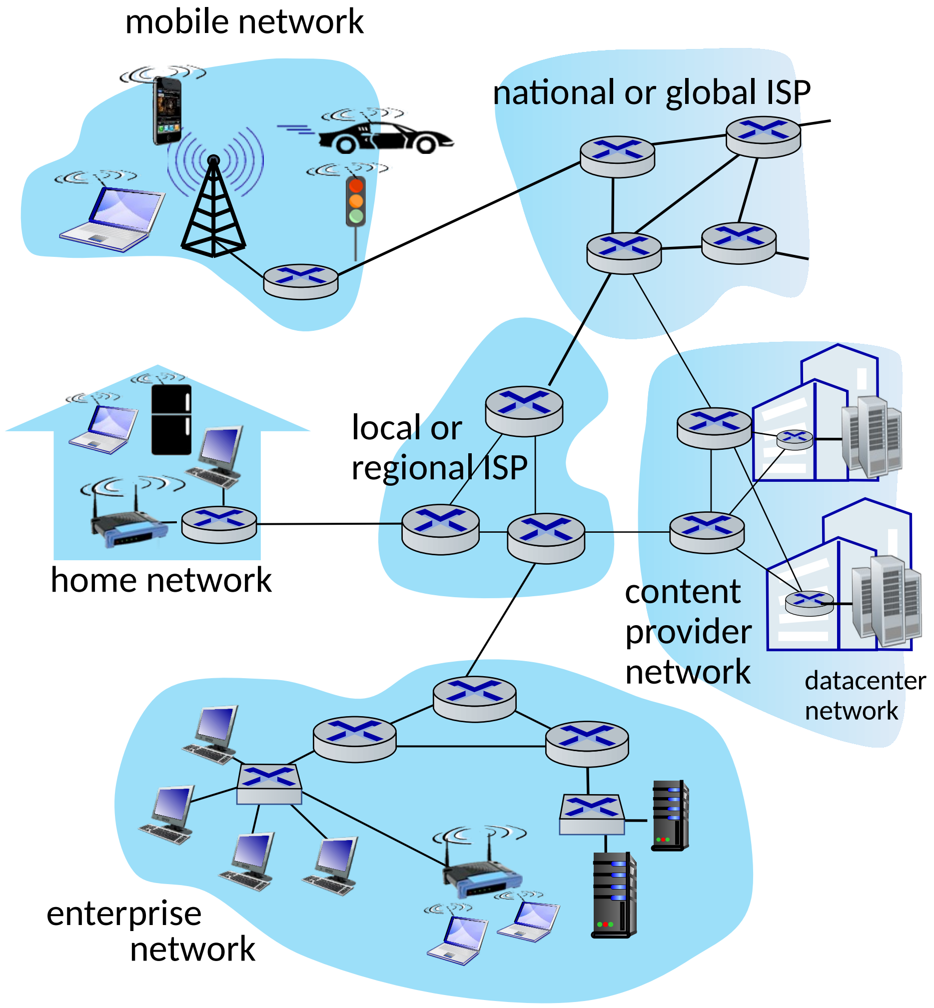
- **Hosts** are devices on which **applications** (or programs) runs:


Laptop


Server


Smartphone


PC

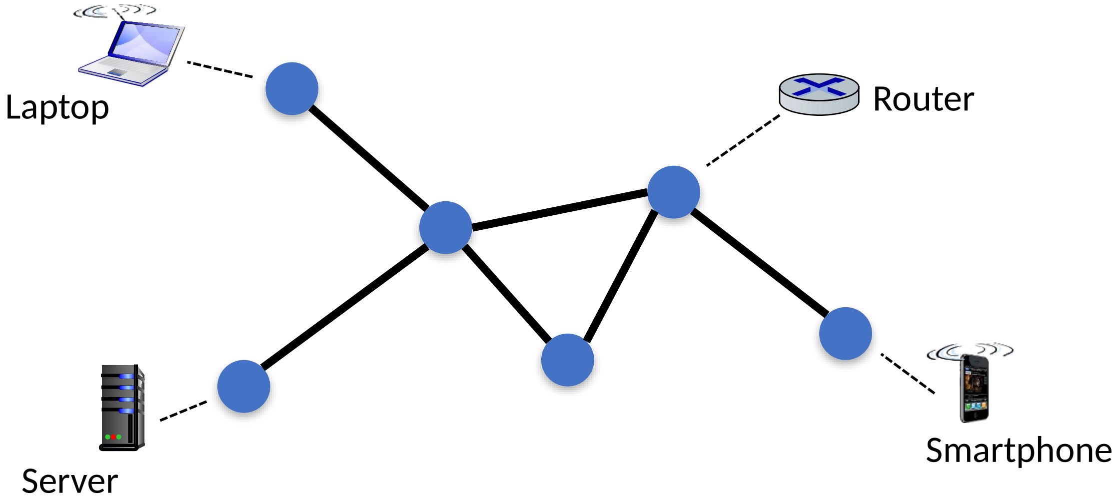


Introduction

Computer Networks and graphs

- Computer networks share terminologies with graph theory:
  - The devices connected through the network are called **nodes** while the connections between nodes are called communication **links** (or **channels**).
  - A **sequence** of nodes/links is called **path**.

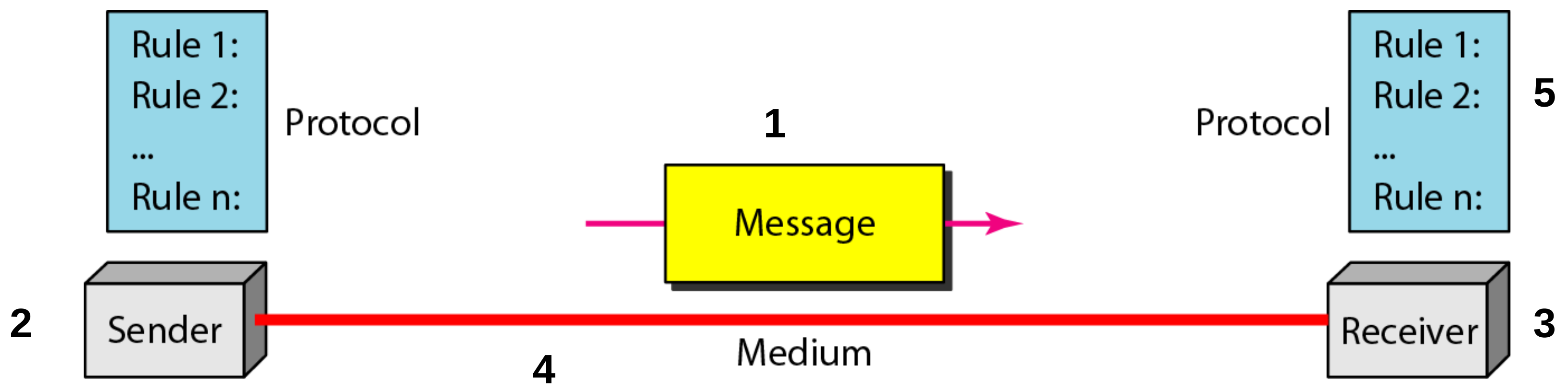
- The end-points (or end-systems) of the network, which provide or use **services**, are special nodes called **hosts**.



Ideally hosts are the leaves of the network (often it is not true) while intermediate nodes are routing devices (routers, switches).

Introduction
Data Communication

- Data communication main components:
  1. **Message**: contains the **data** to be transmitted.
  2. **Sender**: the entity which is sending the message.
  3. **Receiver**: the entity which is supposed to receive the message.
  4. **Medium**: the channel between **sender** and **receiver** where the **message** travels.
  5. **Protocol**: a set of communication rules, known to **sender** and **receiver**.

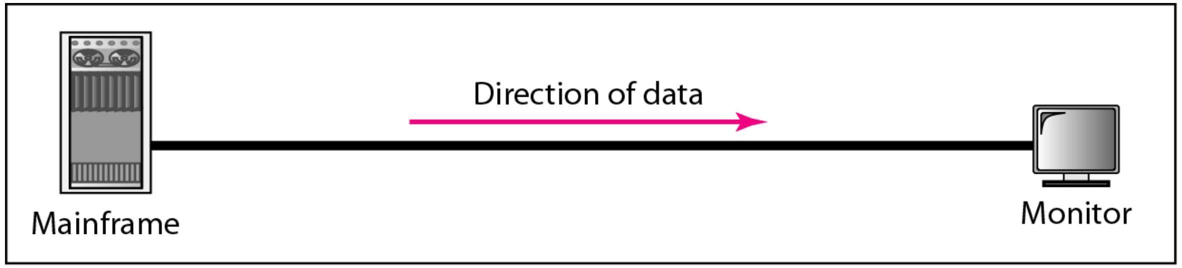


Introduction

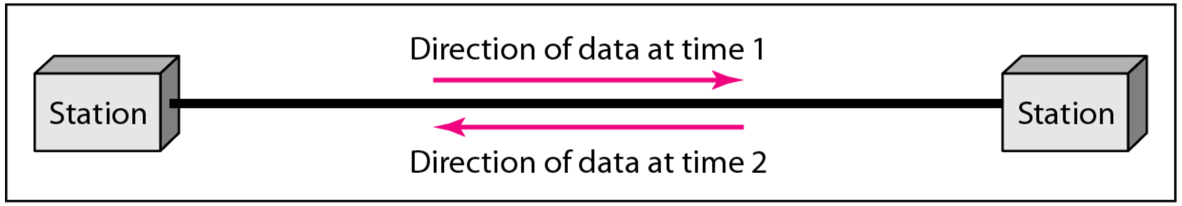
Data Representation and Flow

- The **data** we want to exchange can be represented in different forms.
  - Text, numbers, images, audio, video, etc.

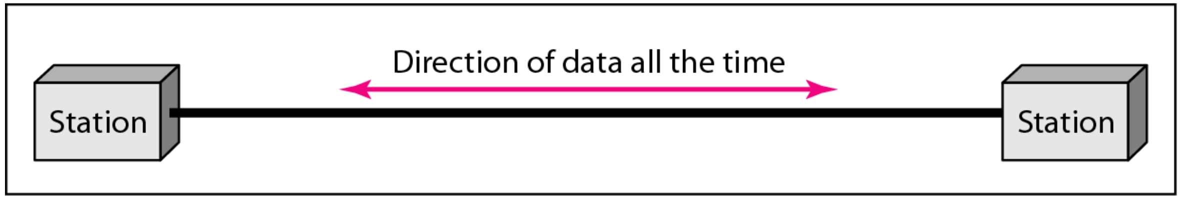
- Depending on the type and the purpose of the communication, the **data flow** can be:
  - **Simplex**: monodirectional.
  - **Half-duplex**: bidirectional (taking turns).
  - **Full-duplex**: bidirectional (simultaneous).



a. Simplex



b. Half-duplex



c. Full-duplex

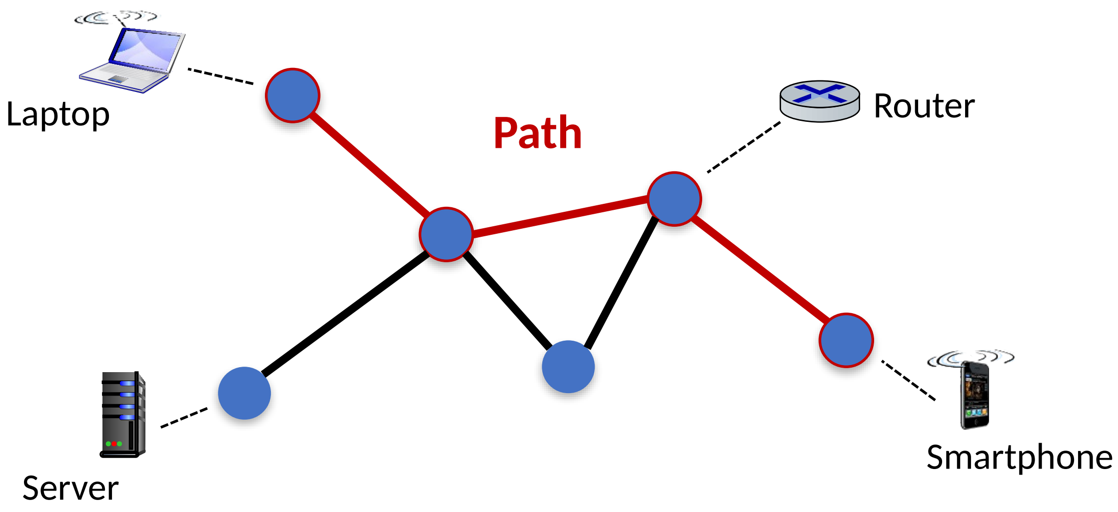
Introduction

Data Representation and Flow

- **Transmission rate** typically refers to the **maximum** amount of information that a **device** can transmit, measured in bit/s (or byte/s).

- The **bandwidth** is the **maximum** amount of information that a **path** (links + nodes) can transmit, measured in **bit/s** (or byte/s).

- The **throughput** is the **actual** (instantaneous) amount of information that is being transmitted over a **path**, measured in **bit/s** (or byte/s).

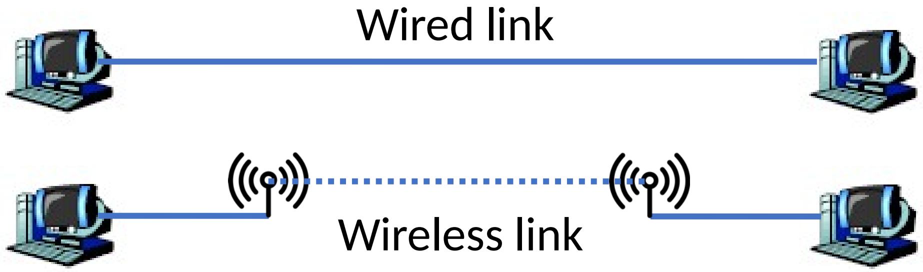


Introduction

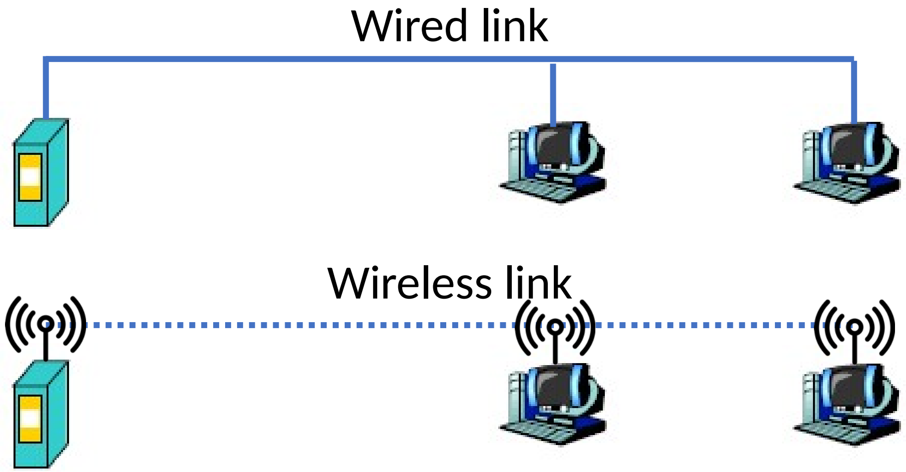
Types of Connections

- The hosts of a network can be connected in different ways.

- **Point-to-point**: a dedicated link is provided between two devices (wireless or wired).



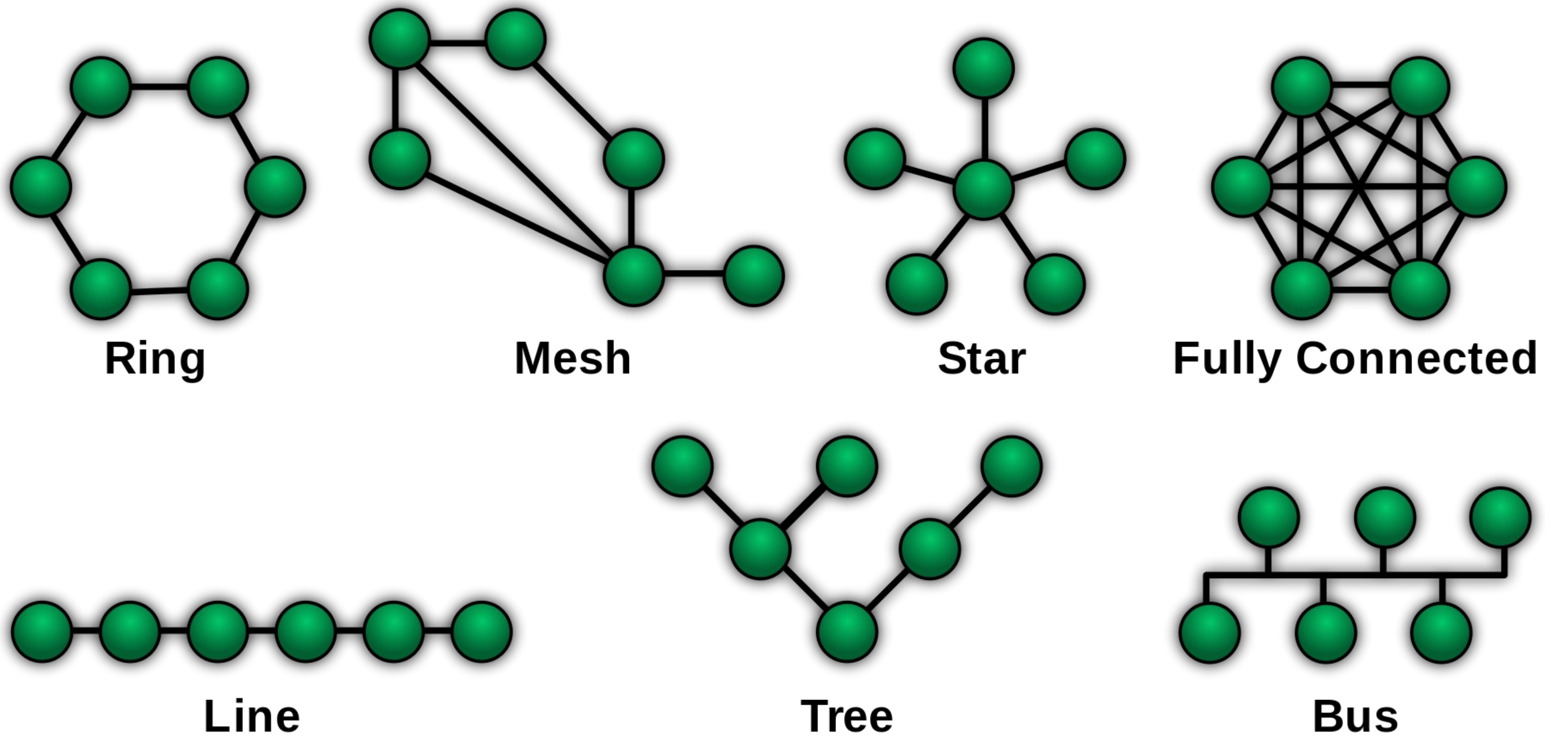
- **Multipoint** (**broadcast**): more than two specific devices share a single link (wireless or wired).



Introduction
Network Topologies

- There are different ways of connecting devices.

- The **network topology** is the arrangement of the elements (links, nodes, etc.) of a communication network.

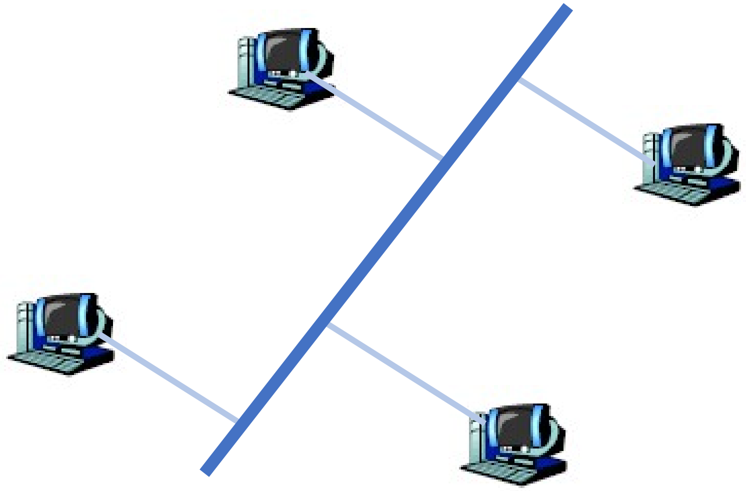


en.wikipedia.org

Introduction

Network Topologies: Bus

- In the **Bus** topology hosts are connected to a central backbone (bus) cable.
  - **Collisions** are possible.



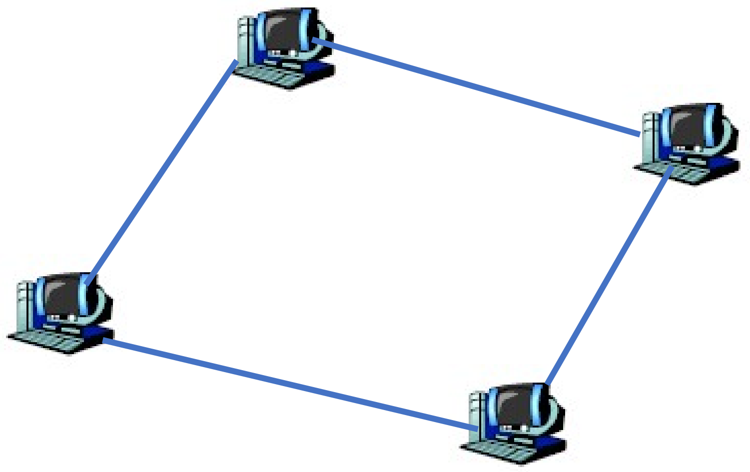
- Pros:
  - Simple and cheap.
  - Good for small networks.
- Cons:
  - Single point of failure (broken bus) but sub networks may still be available.

Bus networks have *1* physical duplex link (or *1* backbone and *n* links).

Introduction

Network Topologies: Ring

- In the **Ring** topology hosts are point-to-point connected to exactly two other ones. The signal is **forwarded** along the ring, from device to device, until it reaches its destination.



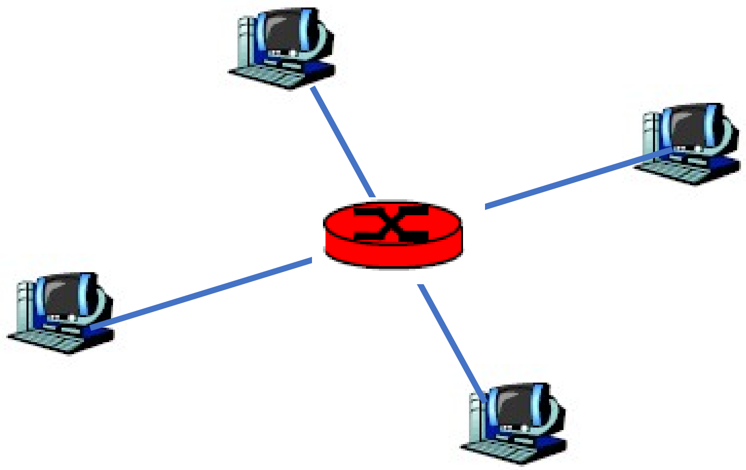
- Pros:
  - Simple and cheap.
  - Performs better than the bus.
- Cons:
  - Adding new nodes is harder.
  - Nodes malfunctioning may impair the network.

Ring networks have *n* duplex links.

Introduction

Network Topologies: Star

- In the **Star** topology hosts are linked to a central controller (hub, switch or router), there is no direct link between hosts.
- The central controller redirects messages.



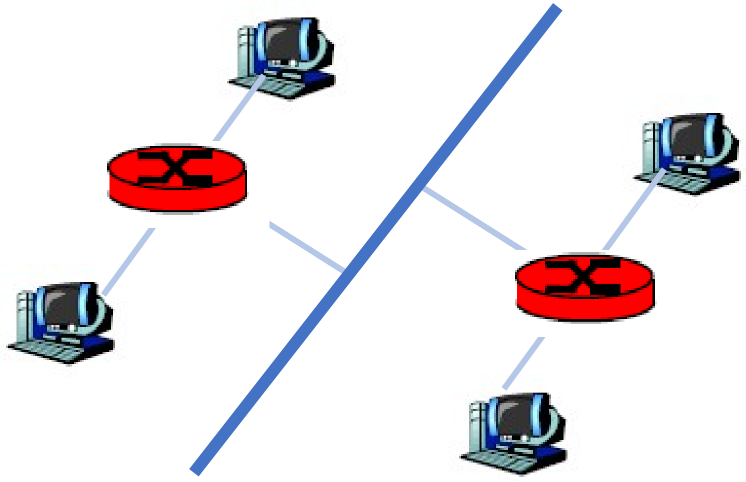
- Pros:
  - Less expensive, simple, robust, more scalable.
- Cons:
  - The controller must be reachable by all hosts.
  - Single point of failure.

Star networks have *1* controller and *n* physical duplex links!

Introduction

Network Topologies: Tree

- In the **Tree** topology multiple star topologies are integrated typically through a bus cable.

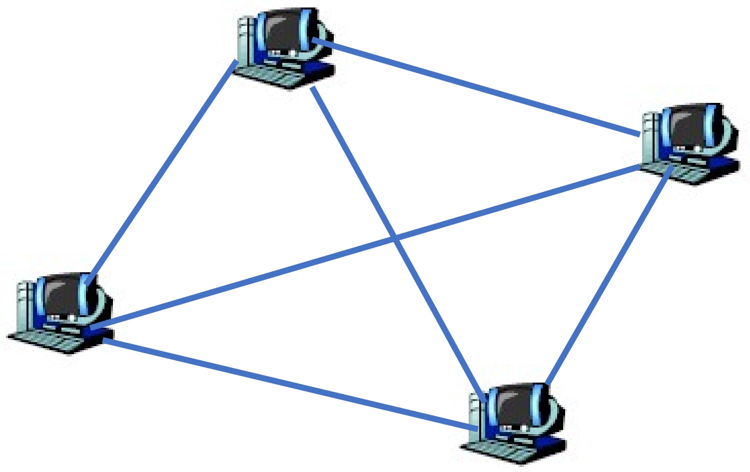


- Pros:
  - Versatile, scalable, robust.
  - Well supported by HW and SW providers.
- Cons:
  - Hard to configure.
  - Weakness of the bus.

Introduction

Network Topologies: Mesh

- In the **Mesh** topology hosts are point-to-point linked in a non-hierarchical way.
  - **Full mesh**: all nodes connected (full-connected).
  - **Partial mesh**: nodes connected with some others.

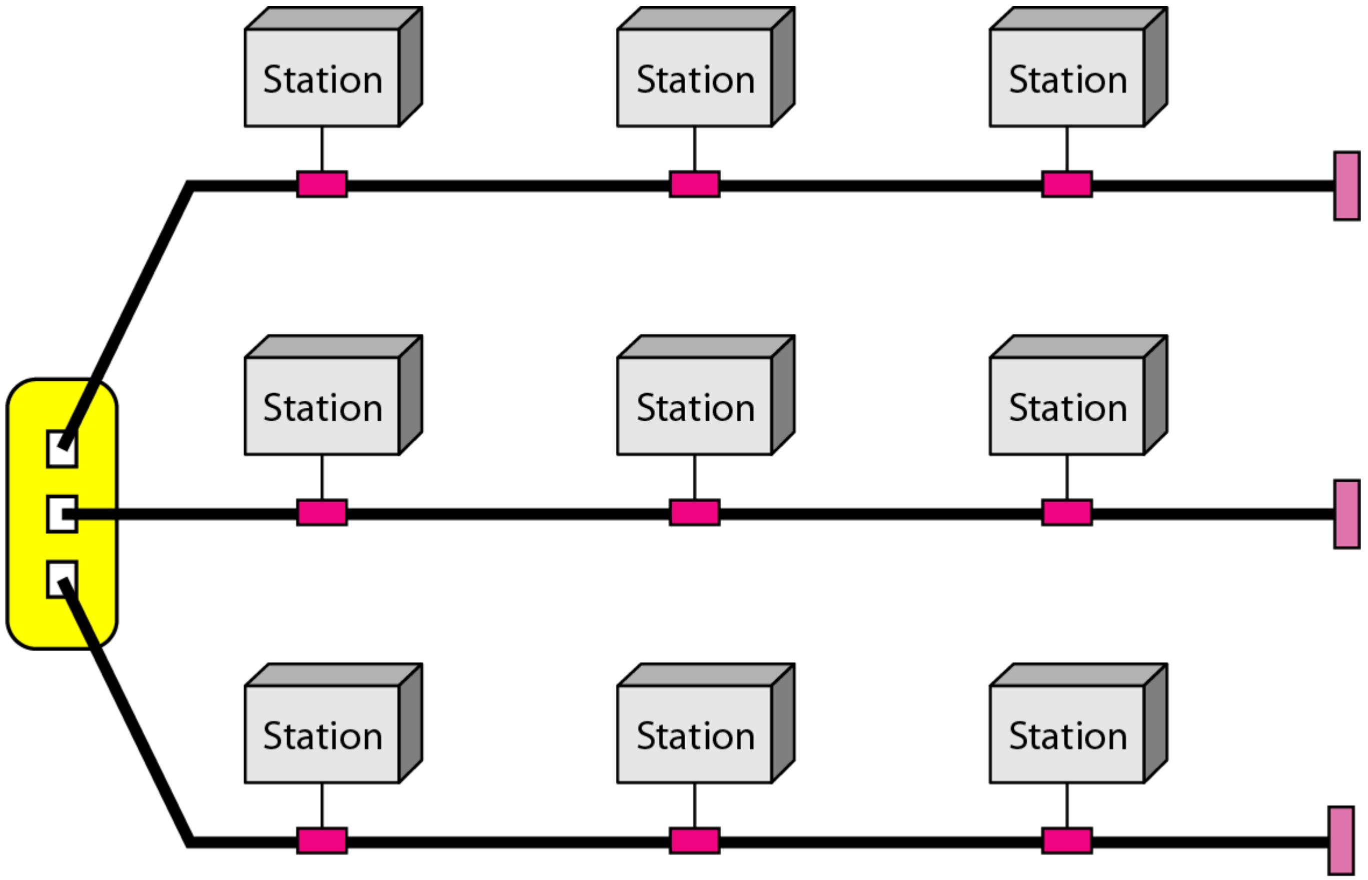


- Pros:
  - Low traffic, robust, secure, dedicated.
- Cons:
  - Hardly scalable.
  - Expansive (need for devices with multiple ports).

Full mesh networks have $n(n-1)/2$ physical duplex links!

Introduction
Network Topologies: Hybrid

- Topologies can be **mixed** into a hybrid network.

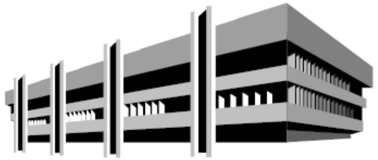


Star-backbone with 3 bus networks

Introduction

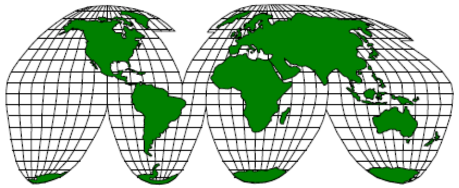
Categories of Networks

- Networks are categorized by size, number of hosts, bandwidth.
  - **Personal Area Network** (PAN)
  - **Local Area Network** (LAN) or Wireless Local Area Network (WLAN).
  - **Metropolitan Area Network** (MAN).
  - **Wide Area Network** (WAN)

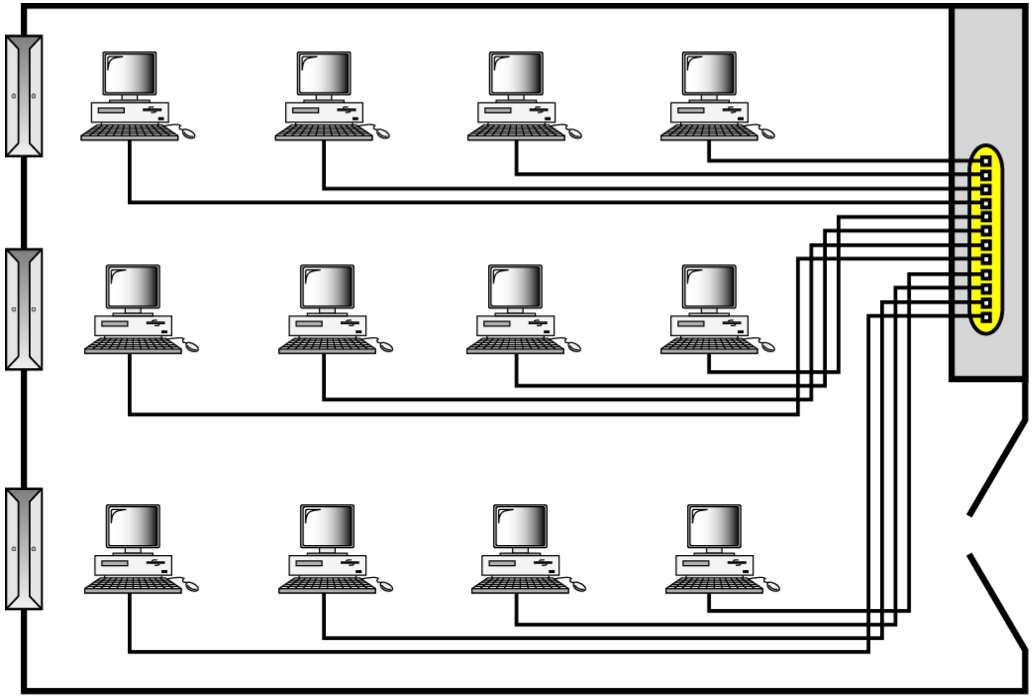


LAN/WLAN

MAN



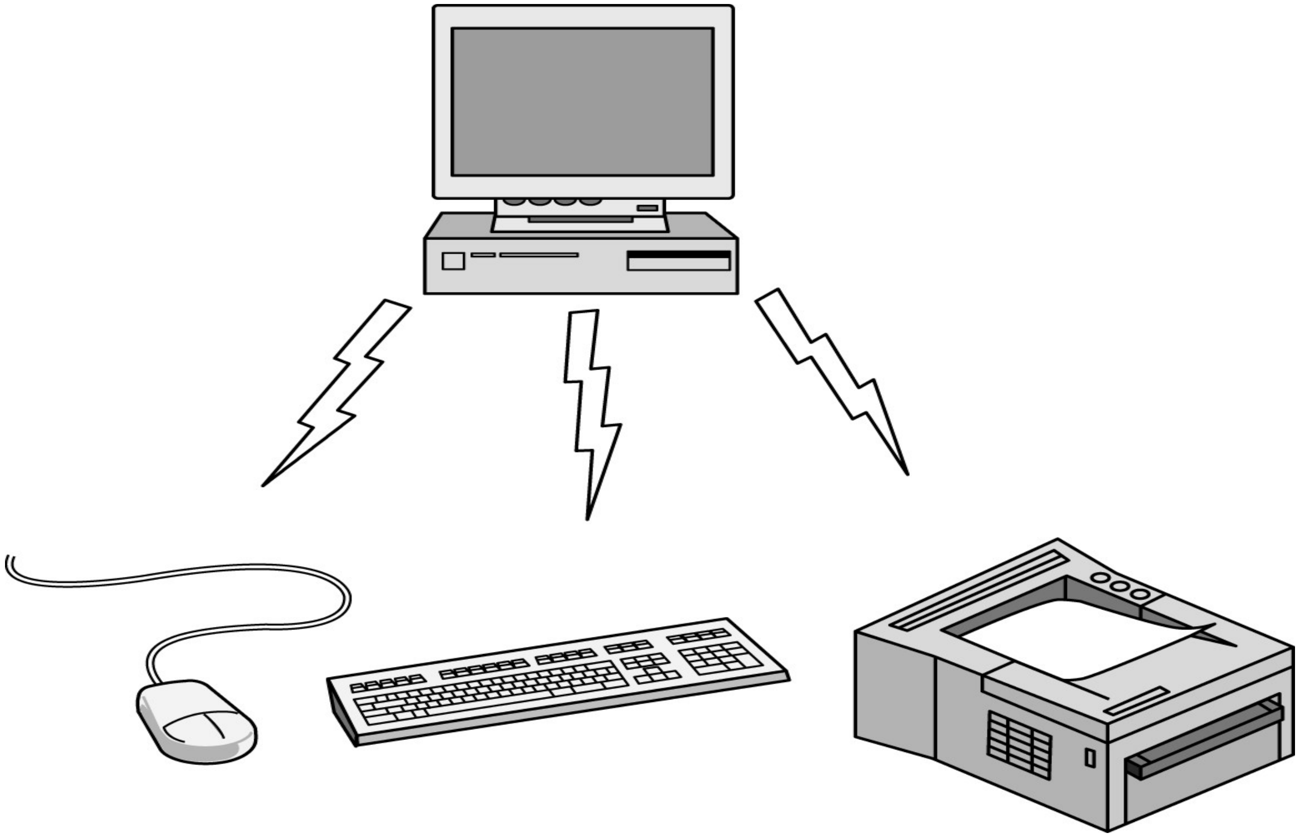
WAN



Example of LAN connecting 12 computers to a switch

Personal Area Network

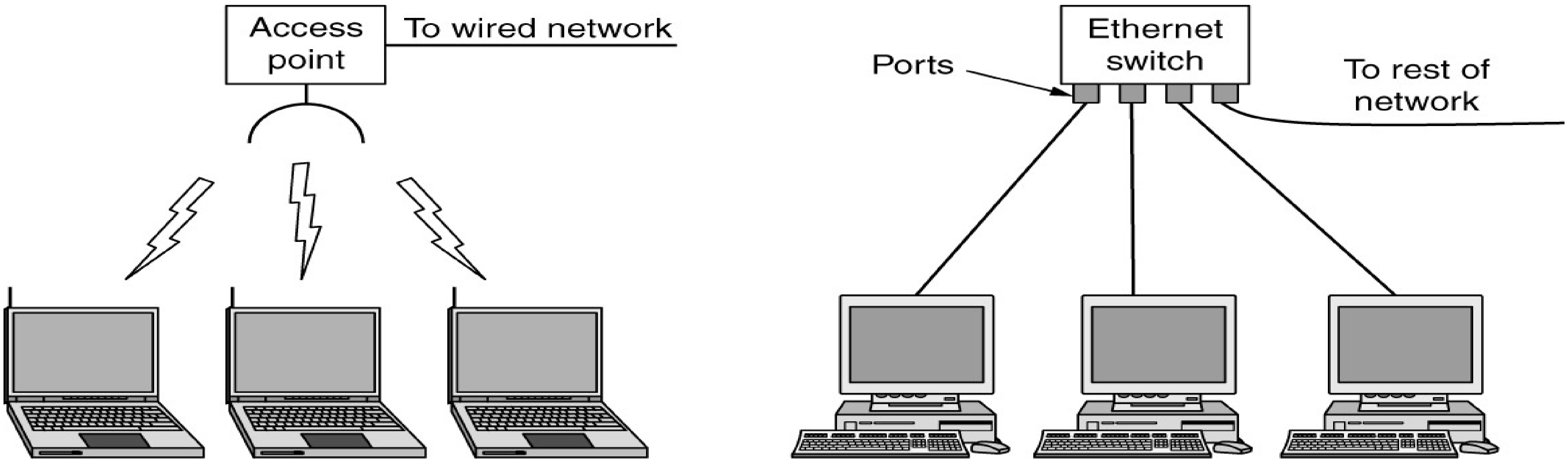
Categories of Networks



PANs (Personal Area Networks) let devices communicate over the range of a person. Bluetooth is a short-range wireless network used to connect components without wires.

Local Area Network

Categories of Networks



The configuration on the left represents a wireless 802.11 network. The configuration on the right represents a wired switched Ethernet network.

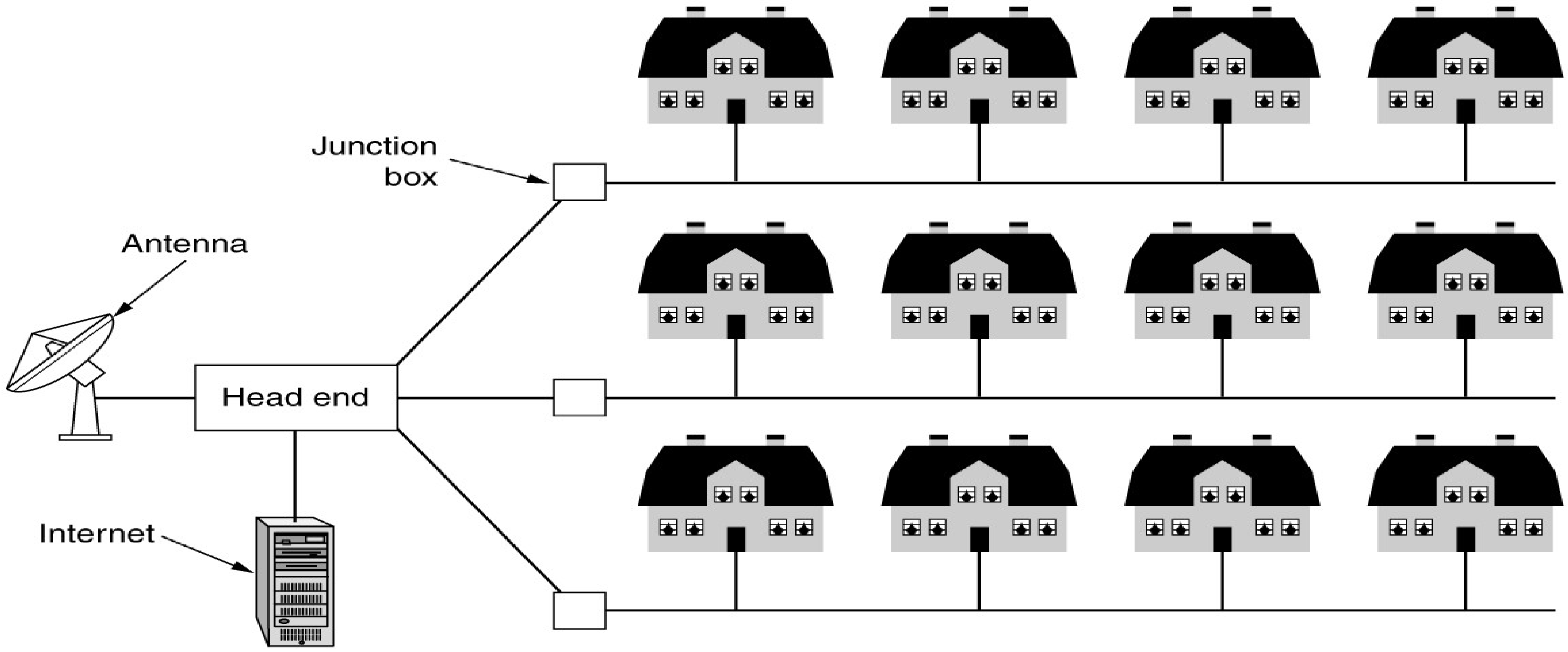
Home Network

Categories of Networks

- Home network LAN
  - Broad, diverse range of Internet-connected devices
  - Characteristics: manageable, dependable, and secure
- Internet of things
  - Allows almost any device to connect
- Required home network properties
  - Easy to install
  - Secure and reliable
  - Interfaces work between all products
  - Reduced consumer device costs

Metropolitan Area Network

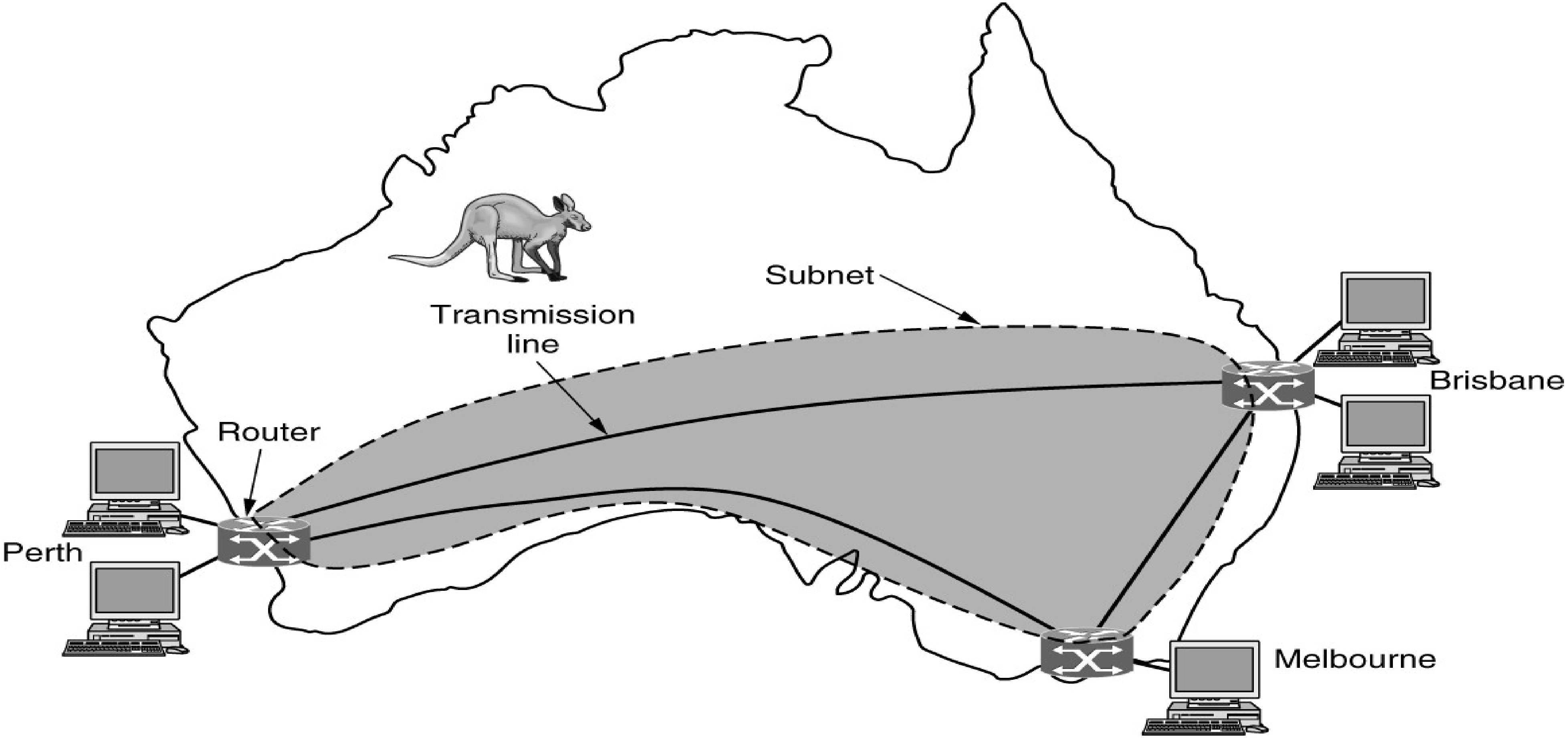
Categories of Networks



A MAN (metropolitan area network) where both television signals and the Internet are being fed into the centralized cable head-end (or cable modem termination system) for subsequent distribution to people’s homes.

Wide Area Network (1/3)

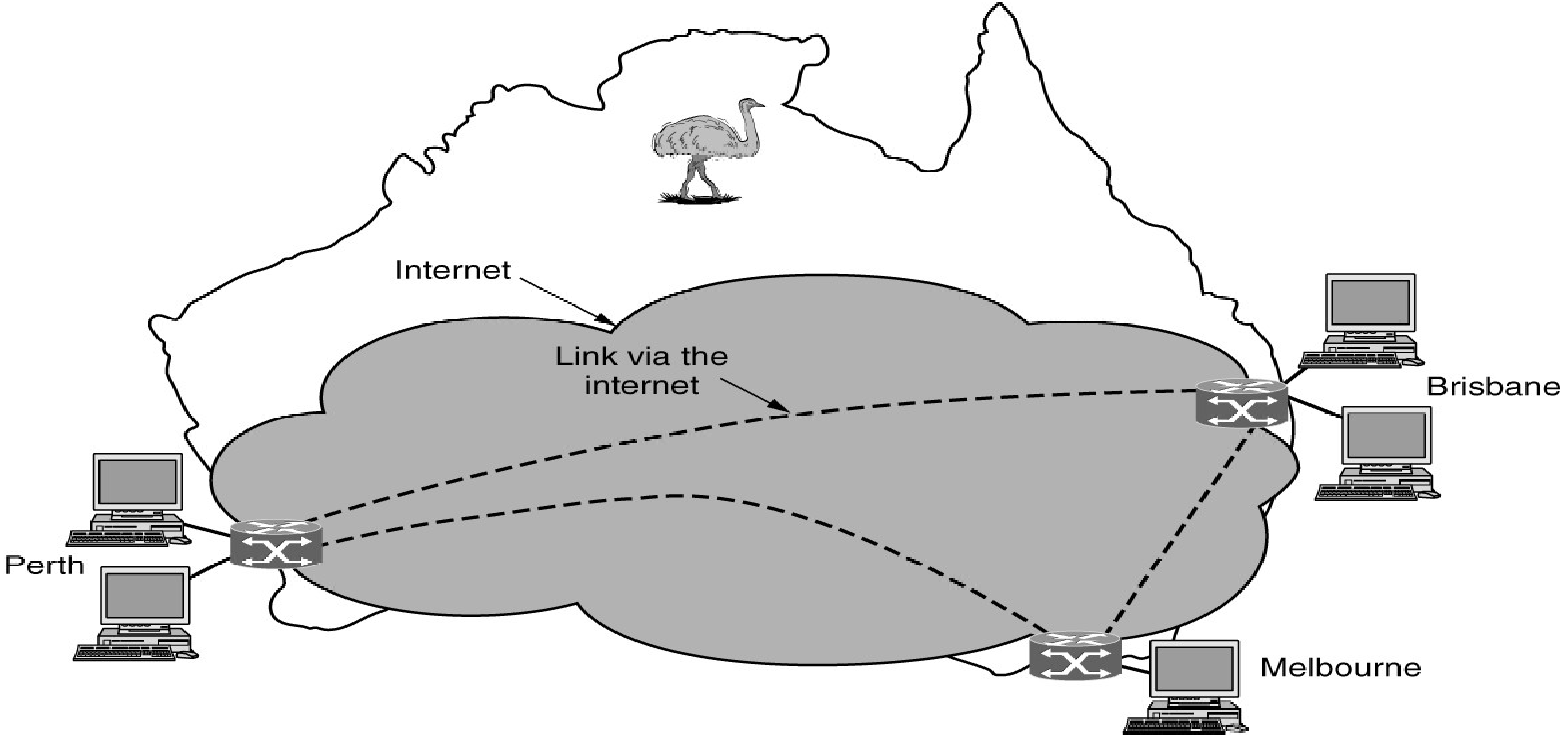
Categories of Networks



This wide area network illustrates how hosts in Perth, Brisbane, and Melbourne can communicate using leased lines.

Wide Area Network (2/3)

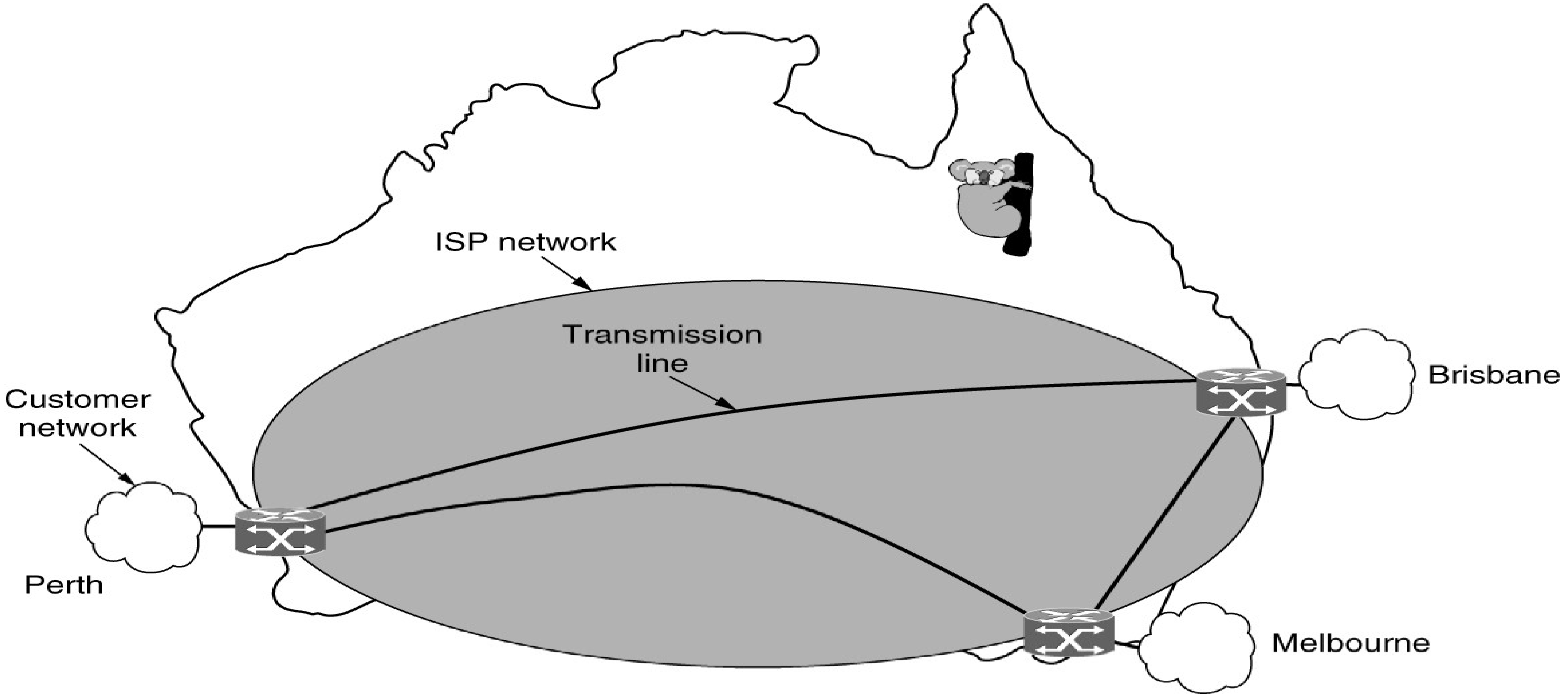
Categories of Networks



This wide area network illustrates how hosts in Perth, Brisbane, and Melbourne can communicate via the Internet.

Wide Area Network (3/3)

Categories of Networks

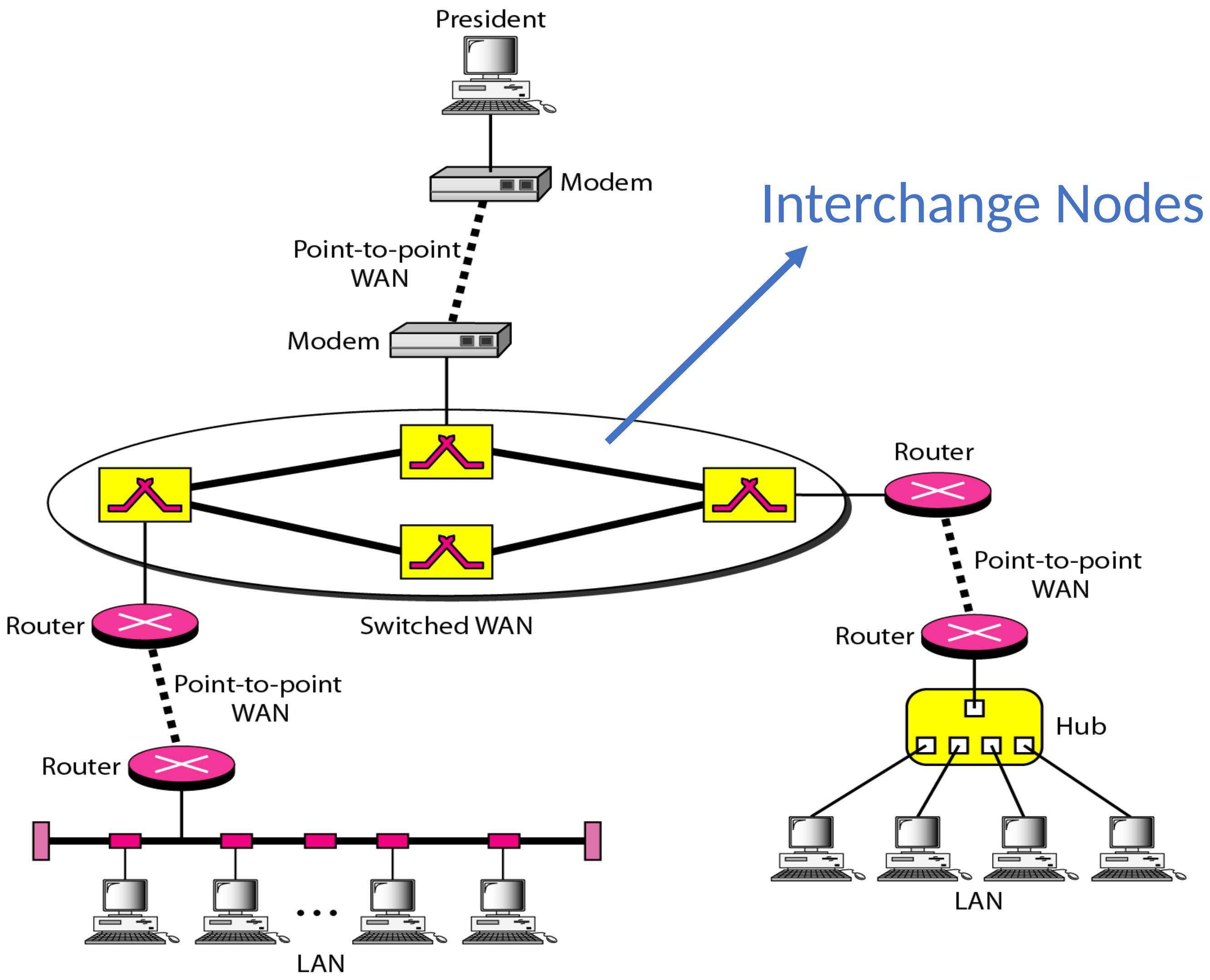


This wide area network illustrates how hosts in Perth, Brisbane, and Melbourne can communicate via an ISP.

Introduction
Complexity of WANs

- The complexity of the topology may increase with the size of the network.

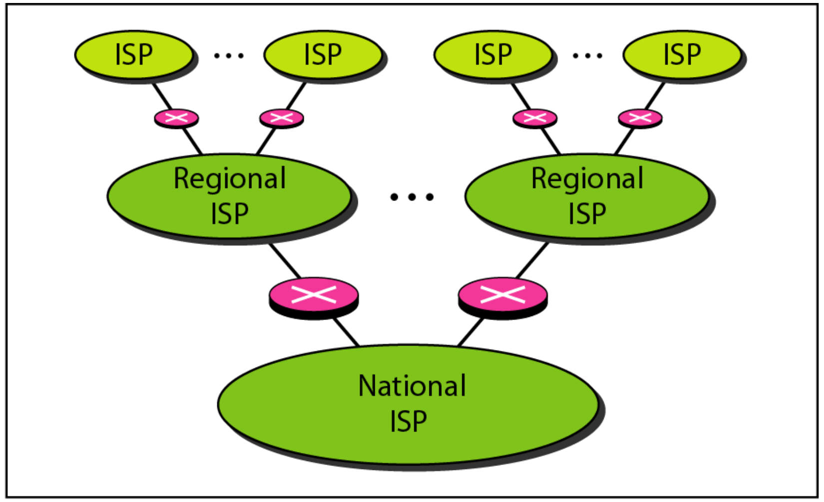
- WANs connecting nations or continents can obviously be very complex and heterogeneous.



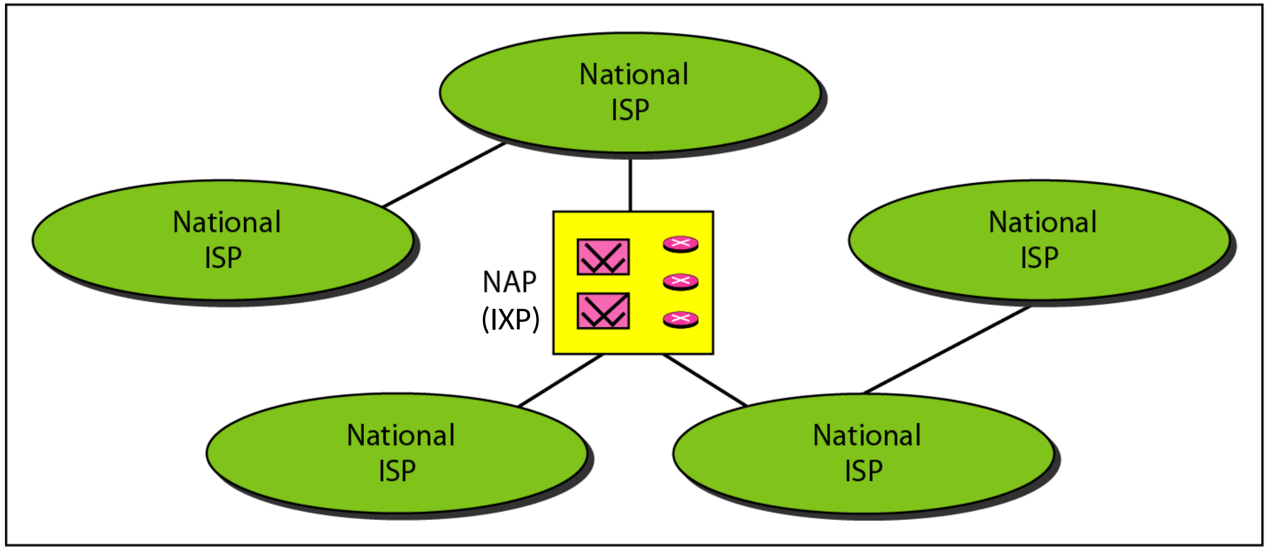
Example of heterogeneous WAN

Introduction
Internet Service Providers

- **Internet access** is the basic service.
- An **Internet Service Provider** (ISP) is an organisation that provides services for accessing, using, or participating the Internet.
- ISPs can be organized as commercial, community-owned, non-profit, or privately owned (e.g., by private companies or universities).
- Different ISPs exchange data through **Network (neutral) Access Points** (NAPs) or **Internet Exchange Points** (IXPs).



a. Structure of a national ISP



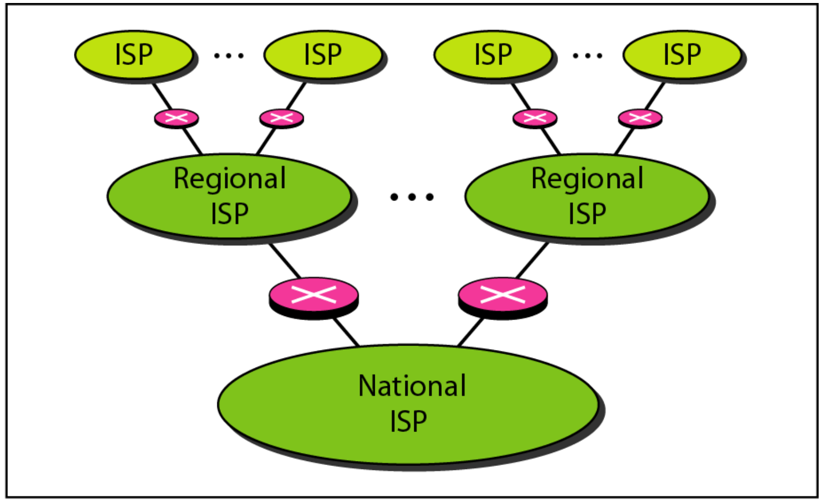
b. Interconnection of national ISPs

Introduction

Internet Service Providers

- An ISP network is hierarchically defined:
  - A **point of presence** (PoP) is a group of one or more routers used by ISPs to reach the customers.
  - **Access** ISPs for local areas.
  - **Regional** ISPs for larger areas.
  - **National** ISPs for nations.

- An **Internet exchange point** (IXP) works as a meeting point between multiple ISPs (often not managed by the ISPs involved).



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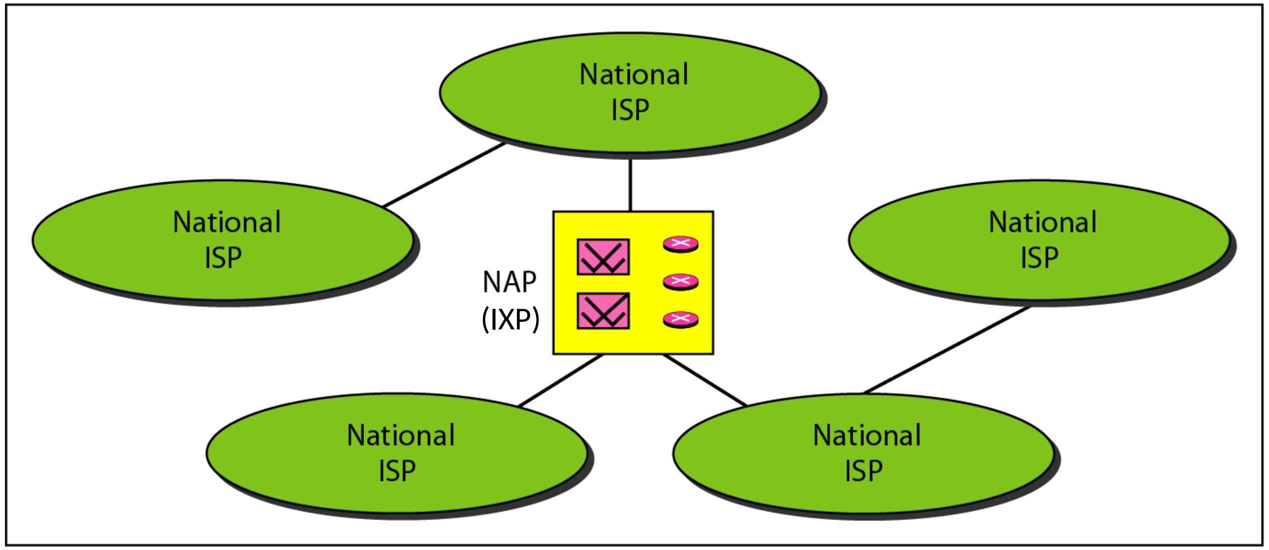

b. Interconnection of national ISPs

Introduction
Internet Connection

- Private internet connections are established via **Modem/ONT** connected to the telephone network.
  - Analogic (56kbps).
  - Integrated Services Digital Network ISDN (128kbps).
  - Asymmetric Digital Subscriber Line ADSL (1Mbps to 20Mbps).
  - Twisted-Pair Copper Wire (10Mbps to 100Mbps).
  - Optical Fiber (50Mbps to 40Gbps), via optical network terminal - ONT, no (de)modulation.

- Companies (especially medium/large ones) can have a direct/dedicated connection to the ISP.